

SRAYAN BHATTACHARYA

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EDUCATION

Institute of Engineering and Management, Kolkata, WB

Jul. 2023 – Jul 2027

Bachelor of Technology in Computer Science & Information Technology

CGPA: 8.59/10 (Through Semester 5)

TECHNICAL SKILLS

Languages: Java, Python, SQL (PostgreSQL), JavaScript, C

Frameworks: Spring Boot, React, Node.js

ML & AI: TensorFlow, PyTorch, Keras

Tools: Git, Docker, Firebase, WordPress

Concepts: REST APIs, WebSockets, Microservices, Machine Learning, IoT Systems

PROJECTS

ReTAQMLN²: Real-Time Air Quality Monitoring Using Liquid Neural Networks

- Engineered an IoT-based air quality monitoring system using Raspberry Pi 5 with SDS011 and BMP280 sensors, capturing **4 real-time environmental parameters** including PM_{2.5} and PM₁₀
- Developed and optimized a Liquid Neural Network, achieving **R² scores of 0.78 (PM_{2.5}) and 0.81 (PM₁₀)** on unseen real-world data (0.90/0.94 during training)
- Processed **10,000+ time-series samples** using normalization, temporal encoding, and sliding window techniques
- Built a real-time dashboard with **sub-second update latency** for live monitoring and forecasting
- Technologies:** Python, PyTorch, Liquid Neural Networks, React, Firebase

CodeDuel: Real-Time Multiplayer Competitive Coding Platform

- Architected a real-time multiplayer coding platform supporting **multiple concurrent users and rooms**
- Implemented WebSocket-based communication achieving **low-latency real-time updates (<200ms)**
- Integrated sandboxed code execution supporting **3+ programming languages**
- Secured platform using JWT authentication handling **role-based access across multiple user types**
- Technologies:** Java, Spring Boot, Spring Security, WebSockets, PostgreSQL, React, TypeScript, Docker

EXPERIENCE

Freelance Web Developer

Jul 2025 – Sep 2025

Remote

- Developed an e-commerce platform with **responsive design across 3+ device types (mobile, tablet, desktop)**
- Improved page load performance by **25–30%** through optimization techniques
- Implemented SEO strategies improving site discoverability and user engagement

RELEVANT COURSEWORK

- Data Structures and Algorithms
- Database Management Systems
- Machine Learning Fundamentals
- Software Engineering